

How to handle LLMs in your research

Pablo Mosteiro, Robert Bagheri, Özgür Togay, Laurence Frank

Partly generated with help of Qwen/Qwen2.5-7B-Instruct and moonshotai/Kimi-K2-Instruct-0905 via together and groq, respectively (HuggingChat).

Department of Methodology & Statistics

June 22, 2026



Universiteit Utrecht

Today's Agenda

- Introduction (3 min)
- How LLMs Work (10 min)
- Limitations and Risks (10 min)
- Prompting CAs and Chatbots (10 min)
- Practical LLM Use in Research (10 min)
- Conclusion (2 min)



About us

Natural Language and Text Processing Lab



<https://nlp.sites.uu.nl/>

Introduction: What do you know about LLMs?

- **Activity: Think-Pair-Share** (2 min)



Image produced in DALL-E December 2023 (Credit: Christine Fox)



Artificial Intelligence, ML, and LLMs: Key Definitions

Artificial Intelligence (AI)

- Computer systems performing tasks requiring human intelligence
- Examples: Chess playing, image recognition, language translation

Machine Learning (ML)

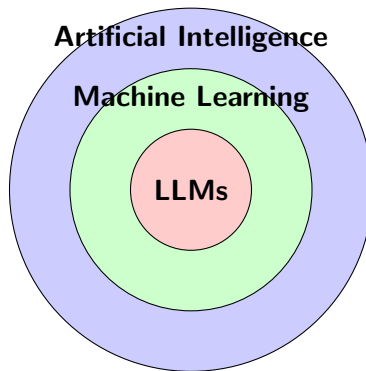
- Subset of AI using algorithms that improve through experience
- Learns patterns from data without explicit programming

Large Language Models (LLMs)

- AI systems trained on vast text amounts
- Understand and generate human language
- Examples: GPT, Claude, Qwen, Mistral



AI vs ML vs LLM: The Relationship



LLMs are a specific type of ML,
which is a subset of AI



Model Learning Process: Training Phase

Training Phase

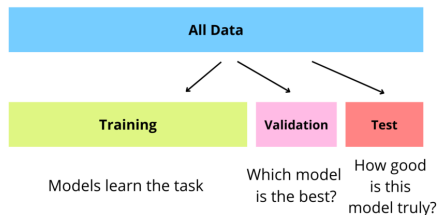
- Model learns patterns from large datasets
- Adjusts model to minimize prediction errors
- Requires massive computational resources
- Can take weeks or months for large models

Example: LLM learns that "cat" often appears with "dog" in pet-related contexts



Model Learning Process: Validation Phase

Validation and Testing



Source: <https://dqlab.id/5-panduan-praktis-membuat-model-machine-learning>

Key insight: Good models **generalize**, not just **memorize**



Universiteit Utrecht

Prompting phase

U weet wat ik ga zeggen

Credit for this slide: Albert Gatt & Dong Nguyen



Universiteit Utrecht

U weet wat ik ga



What is Generative AI?

Definition: AI that creates new content (text, images, code, audio)

Key Features:

- Creates novel outputs, not just classifies
- Learns patterns from training data
- Generates human-like content

Examples: ChatGPT (text), DALL-E (images), CodeT5 (code)



Text-GenAI components

	consists of	fed by	GPT learns
	LLM	textual internet data	“knowledge” of language and content
	Dialogue Model	social media, forums	“knowledge” of dialogue structure
	Task Model	programmers	tasks, self knowledge and limits
	Ethical Governor	Kenian data labellers	ethical answers

Credit for this slide: Davitze Könning

Reinforcement Learning



Universiteit Utrecht

Generative AI vs. Other Models

Feature	Generative	Discriminative
Primary Task	?	?
Output Type	?	?
Example	?	?



Generative AI vs. Other Models

Feature	Generative	Discriminative
Primary Task	Create new content	Classify existing data
Output Type	Novel text/images	Labels/categories
Example	GPT, DALL-E	Spam filters, medical diagnosis



Activity: True or False?

Instructions: Raise your hand if you think the statement is TRUE

- **Statement 1:** Generative AI can only work with text
- **Statement 2:** All generative models are also discriminative

Caveat

gold labels!

- **Statement 3:** Generative AI creates completely original content never seen in training
- **Statement 4:** LLMs are a type of generative AI



Limitations and Risks: Hallucinations

What are hallucinations?

- LLMs generate false or nonsensical information
- Appears confident and plausible
- Can include fake citations, non-existent facts



Bias

- **Gender bias:** Assumes doctors are male, nurses are female
- **Racial bias:** Associates certain groups with negative stereotypes
- **Cultural bias:** Primarily represents Western perspectives
- **Academic bias:** Favors published, English-language sources
- **Representation bias:** Underrepresents certain groups
- **Confirmation bias:** Reinforces existing beliefs
- **Temporal bias:** Becomes outdated as language/society changes

Real-world impact: Biased models can perpetuate harmful stereotypes



Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers

Give me the answer to the question "What is life?"

The question "What is life?" is a profound and complex one that has been explored by philosophers, scientists, theologians, and thinkers throughout history. There isn't a single, universally accepted answer, but here are some perspectives:

Give me the answer to the question "What is life?"

The question "What is life?" is a profound and complex one that has been explored by various disciplines, including biology, philosophy, theology, and more. Here are some key perspectives:

- **Context dependency:** Performance varies by domain

Pre-Chorus:

Limitations and Risks: Ethics

Ethical Concerns

- **Data:** Training data may include personal or copyrighted material
- **Environmental impact:** High energy consumption
- **Labor practices:** Data annotation often involves poor working conditions



"When it gets to the day before payday, they close the account and say that you violated a policy," Kanyugi said.

Employees say they have no recourse or even a way to complain.

The company told 60 Minutes that any work done "in line with our community guidelines was paid out." In March, as workers started complaining publicly, Remotasks abruptly shut down in Kenya, locking all workers out of their accounts.

The mental toll of AI training

Workers say some of the projects for Meta and OpenAI also caused them mental harm. Wambalo was assigned to train AI to recognize and weed out pornography, hate speech and excessive violence from social media. He had to sift through the worst of the worst content online for hours on end.

"I looked at people being slaughtered," Wambalo said. "People engaging in sexual activity with animals. People abusing children physically, sexually. People committing suicide."

Berhane Gebrekidan thought she'd been hired for a translation job, but she said what she ended up doing was reviewing content featuring dismembered bodies and drone attack victims.

<https://www.cbsnews.com/news/ai-work-kenya-exploitation-60-minutes> (Nov 2024)

- **Academic integrity:** attribution of AI assistance / uploading restricted data
- **Open source:** Often little documentation on data, training, architecture

Effective Prompting: BRAVE(R) Framework

	B: Boundaries	Set limits on the format, length, or any other constraints.
	R: Role	Identify the role or perspective you want the AI tool to take.
	A: Audience	Specify who the output is intended for to determine the appropriate tone and style.
	V: Variables	Highlight key details, variables, or points that should be included in the response.
	E: Expectations	Clearly state what you expect the AI to accomplish, including the intended outcome and purpose.
	(R): Refine	Provide feedback to improve the output and guide future interactions.

Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

Poor prompt: “Summarize this paper”

Better prompt: “Act as a psychology research assistant [R]. Summarize this paper in 3 bullet points [B] focusing on methodology and findings relevant to cognitive behavioral therapy [V]. Use academic language [A] and APA style [E].”

Key insight: Specific, contextual prompts yield better results

Evaluating and Critically Assessing GenAI Responses

Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).



F: Focus

Focus on the AI's response by breaking it down into its main points and arguments.



A: Authenticate

Authenticate the AI's output by verifying it against current research and empirical data.



C: Critique

Critique the response's accuracy, relevance, and depth in the context of the topic.



T: Think

Think about the response from various perspectives and its potential impact.



S: Scrutinise

Scrutinise the AI's conclusions by critically questioning and exploring alternative hypotheses or interpretations.

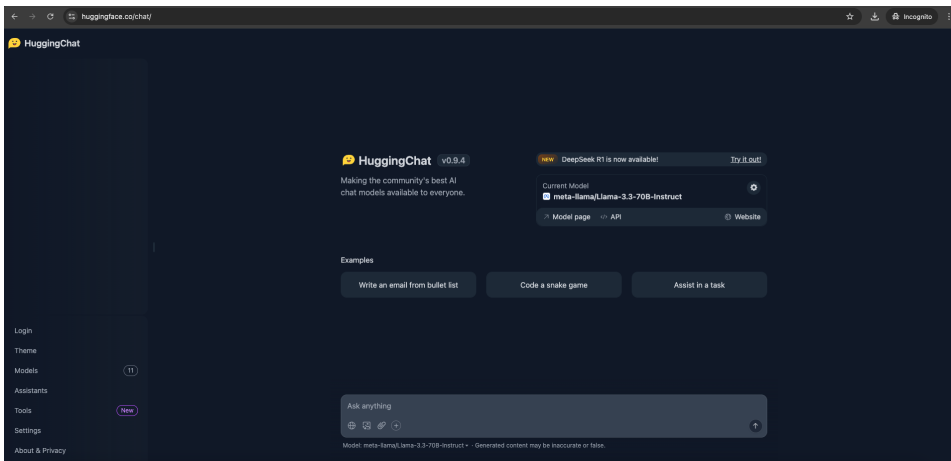
More info:

<https://teachersguidegsls.nl/genai/generative-ai-guidelines/>



Universiteit Utrecht

Ways to prompt



Switch models

×

Models

meta-llama/Llama-3.3-70B-Instruct **Active**

Qwen/Qwen2.5-72B-Instruct

CohereForAI/c4ai-command-r-plus-08-2024

deepseek-ai/DeepSeek-R1-Distill-Qwen-32B

nvidia/Llama-3.1-Nemotron-70B-Instruct-HF

Qwen/QwQ-32B-Preview

Qwen/Qwen2.5-Coder-32B-Instruct

meta-llama/Llama-3.2-11B-Vision-Instruct

NousResearch/Hermes-3-Llama-3.1-8B

mistralai/Mistral-Nemo-Instruct-2407

microsoft/Phi-3.5-mini-instruct

Assistants

My Assistants

meta-llama/Llama-3.3-70B-Instruct

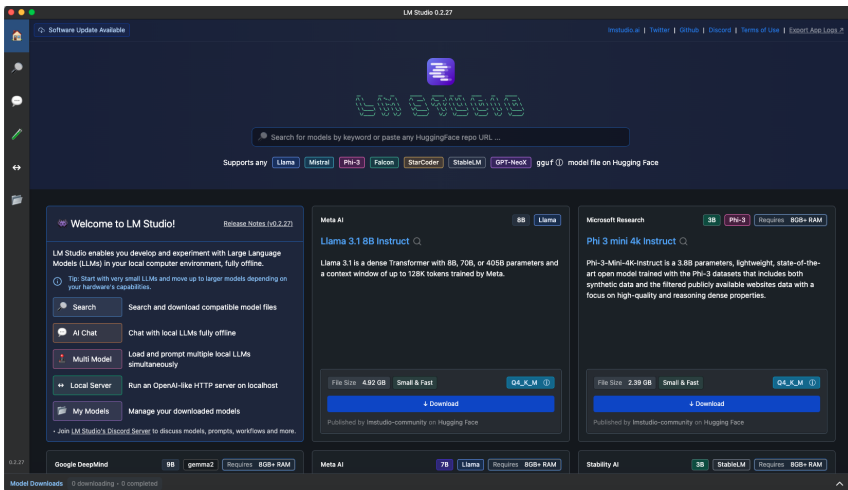
Ideal for everyday use. A fast and extremely capable model matching closed source models' capabilities. Now with the latest Llama 3.3 weights!

[Model page](#) [Model website](#) [API Playground](#) [Copy direct link to model](#)

New chat

System Prompt

Using an LLM locally (no server)



Official UU policy

<https://www.uu.nl/en/organisation/ai-policy/researchers>

- Benefits and use cases
- Risks and limitations
- Effective use of GenAI (*Background check*)
- Citing GenAI in your work



LLM Use in Research

Appropriate & Inappropriate Uses

- Think-Pair-Share



LLM Use in Research: Appropriate Applications

Appropriate Uses

- **Literature review:** Summarizing papers (with verification)
- **Writing assistance:** Drafting, editing, proofreading
- **Code generation:** Writing analysis scripts

Caveat

only when you know which statistical analysis is suitable

- **Brainstorming:** Generating research ideas
- **Translation:** Working with multilingual sources

Key: Always maintain human oversight and verification



LLM Use in Research: Inappropriate Applications

Avoid Using LLMs For:

- **Statistical analysis:** LLMs can't perform proper statistical tests
- **Fact-checking:** They may hallucinate information
- **Critical decisions:** Participant safety, ethical approvals
- **Original thinking:** Generating truly novel theories
- **Replacing expertise:** They supplement but don't replace researcher knowledge



Working with others

nature

Explore content ▾ About the journal ▾ Publish with us ▾ Subscribe

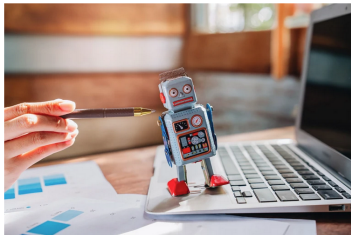
[nature](#) > [career column](#) > [article](#)

CAREER COLUMN | 22 April 2026

Don't let your students use AI as a ghostwriter

How a research proposal generated by artificial intelligence transformed my approach to teaching and supervision.

By [Yanjun Shen](#)



Using generative artificial intelligence models to kick ideas around is more effective than outsourcing thinking wholesale, says Yanjun Shen. Credit: iabkitticha/Stock via Getty

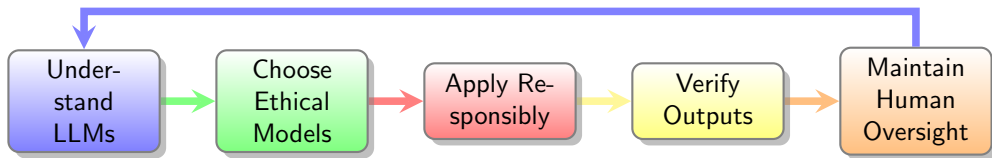
- ① thinking for yourself
- ② read literature
- ③ ask AI for advice
- ④ critically analyze

What we've done

- How LLMs Work
- Limitations and Risks
- Prompting CAs and Chatbots
- Practical LLM Use in Research



Key Takeaways: LLM Workflow



Remember:

- LLMs are powerful tools, not replacements for human expertise
- Always prioritize ethics, privacy, and transparency
- Verify outputs and maintain critical thinking

